

DR. NICOLÁS BUDINI

PERSONAL INFORMATION

Date of Birth: December 15, 1981
Place of Birth: Paraná, Entre Ríos, ARGENTINA
National ID: 29121394
Nationality: Argentine
Address: Cura Álvarez 90 6B (CP E3100XAI)
Paraná, Entre Ríos, ARGENTINA
Email: budinense@gmail.com

PROFESSIONAL CONTACT INFORMATION

Semiconductor Physics Group
Instituto de Física del Litoral (IFIS Litoral, UNL-CONICET), Argentina
Güemes 3450 (CP S3000GLN)
Santa Fe, ARGENTINA
Phone: (+54)342-4559174 ext. 2174
Email: nicolas.budini@santafe-conicet.gov.ar
Department of Physics
Facultad de Ingeniería Química, Universidad Nacional del Litoral
Sgo. del Estero 2829 (CP S3000AOM)
Santa Fe, ARGENTINA
Phone: (+54)342-4571164 - ext. 2562 (then 36)
Email: nbudini@fiq.unl.edu.ar

CURRENT RESEARCH TOPICS

Optics, Digital Holography, Digital Holographic Interferometry, Digital Holographic Microscopy, Dynamic Speckle, Optical Characterization of Materials, Microfluidics in Porous Silicon / Physics Education / Artificial Intelligence & Machine Learning / Meteorology

[Google Scholar Research Profile](#)

EDUCATION

Postgraduate Degree: Ph.D. in Physics

- Graduation Date: March 2012
- Institution: Universidad Nacional del Litoral
- Thesis Title: *Polycrystalline Silicon for Photovoltaic Devices*
- Advisor/Co-Advisor: Dr. Roberto D. Arce / Dr. Javier A. Schmidt
- Field of Study: *Semiconductors, Photovoltaic Devices*

CURRENT EMPLOYMENT

Instituto de Física del Litoral (IFIS Litoral, CONICET-UNL) – Santa Fe
Semiconductor Physics Group

Staff Researcher (CONICET), since November 2017

Universidad Nacional del Litoral (UNL) – Santa Fe
Facultad de Ingeniería Química (FIQ), Department of Physics

Full Professor, since August 2025 (UNL Resolution CS 466/25). *Category IV* in the National University Incentive Program for Researcher Professors (*Category I* requested in the PRINUAR 2023 call, not resolved yet).

- Courses: *Physics I, Quantum Mechanics I & II, Experimental Physics III, Digital Holographic Microscopy*

Director of the Physics Department at FIQ, since April 2022 (FIQ Resolution CD 147/22)

POSTGRADUATE COURSES

1. Quantum Mechanics

- Role: Lead Instructor
- Program: Master in Physics and related fields (Res. CD 1558/22 FBCB, UNL)
- Institution: Facultad de Bioquímica y Ciencias Biológicas / Facultad de Ingeniería Química (UNL)
- Period: since December 2022

2. Digital Holographic Microscopy

- Role: Lead Instructor
- Program: Master and Ph.D. in Physics and related fields (Res. CD 673/24 FBCB, UNL)
- Institution: Facultad de Bioquímica y Ciencias Biológicas / Facultad de Ingeniería Química (UNL)
- Period: since August 2024

3. Simulation Models

- Role: Lead Instructor
- Program: Master in Research Methodology (Res. CD 447/16 FCECO, UNER)
- Institution: Facultad de Ciencias Económicas (UNER)
- Period: since October 2016

LATEST SCIENTIFIC PUBLICATIONS (SINCE 2020)

1. E. Vicentin, N. Cabana, J. L. Navarro Sánchez, R. Nazmutdinov, N. Budini, P. Quaino. *Pt/ γ -graphyne as a new electrode material for fuel cells*. Applied Surface Science **722** (2026) 165553. DOI: [10.1016/j.apsusc.2025.165553](https://doi.org/10.1016/j.apsusc.2025.165553)
2. L. G. Cenchá, A. Benítez, N. Budini, C. L. A. Berli, R. Urteaga. *Directional Drying in Bilayer Porous Films: Funnel vs Ink-Bottle Geometries*. Langmuir **41**(38) (2025) 26044–26053. DOI: [10.1021/acs.langmuir.5c02674](https://doi.org/10.1021/acs.langmuir.5c02674)
3. A. C. Monaldi, J. I. Díaz, F. Martínez, N. Budini, W. A. Báez. *Digital Holographic Microscopy and Machine Learning for Quantitative 3D Analysis and Automatic Classification of Volcanic Ash Particles*. Journal of Geophysical Research - Atmospheres **130**(13) (2025) e2024JD043283. DOI: [10.1029/2024JD043283](https://doi.org/10.1029/2024JD043283)
4. M. Sallese, N. Budini, P. Psota, R. Urteaga. *Three-dimensional characterization of nanoporous membranes by capillary filling using high speed interferometry*. Review of Scientific Instruments **96**(1) (2025) 013901. DOI: [10.1063/5.0223997](https://doi.org/10.1063/5.0223997)
5. M. F. Martínez, M. D. Sallese, P. Psota, C. L. A. Berli, R. Urteaga, N. Budini, A. C. Monaldi. *Filling fraction measurement around a drying drop onto nanoporous silicon using digital holographic microscopy*. Journal of Applied Physics **135** (2024) 073102. DOI: [10.1063/5.0190518](https://doi.org/10.1063/5.0190518)
6. L. G. Cenchá, G. F. García, N. Budini, R. Urteaga, C. L. A. Berli. *Time-temperature indicator based on the variation of the optical response of photonic crystals upon polymer infiltration*. Sensors and Actuators A: Physical **341** (2022) 113571. DOI: [10.1016/j.sna.2022.113571](https://doi.org/10.1016/j.sna.2022.113571)
7. N. Budini, M. Sallese, L. Cenchá, C. L. A. Berli, R. Urteaga. *Implementation of digital holographic microscopy for capillary filling measurements in nanoporous materials*. Applied Optics **61**(10) (2022) 2506-2512. DOI: [10.1364/AO.450570](https://doi.org/10.1364/AO.450570)
8. M. Sallese, J. Torga, E. Morel, N. Budini, R. Urteaga. *Optical coherence tomography measurement of capillary filling in porous silicon*. J. Appl. Phys. textbf128 (2020) 024701. DOI: [10.1063/1.5145270](https://doi.org/10.1063/1.5145270)

9. G. D. Pereyra, F. Y. Oliva, N. Budini, G. Risso, P. D. Pérez, S. Suárez, J. C. Trin-cavelli. *Standardless determination of nanometric thicknesses in stratified samples by electron probe microanalysis*. Spectrochim. Acta B **171** (2020) 105932. DOI: [10.1016/j.sab.2020.105932](https://doi.org/10.1016/j.sab.2020.105932)
10. O. Marín, V. González, N. Budini, M. Tirado, D. Comedi. *The influence of methanol and NH₄Cl on solvothermal ZnO synthesis and properties*. Appl. Phys. A **126** (2020) 466. DOI: [10.1007/s00339-020-03636-w](https://doi.org/10.1007/s00339-020-03636-w)

TECHNOLOGICAL DEVELOPMENTS

Invention Patents

- Un procedimiento para fabricar dispositivos microfluídicos (Process for manufacturing microfluidic devices)
 - Authors: L. Cenchá, R. Urteaga, N. Budini, C. L. A. Berli
 - Granted: September 2023 (Exp. 2037)
 - Argentine Patent No.: AR109785B1
 - Assignees: CONICET - Universidad Nacional del Litoral, Santa Fe, Argentina

PARTICIPATION IN RESEARCH PROJECTS

As Director

1. CAI+D UNL 85520240100105LI: *Weather prediction based on artificial intelligence and development of a lightning detector*, 2025-2028 (project admitted in September 2024).
2. CAI+D Joven UNL 50020150100027LI: *Implementation of digital holographic techniques for the characterization of nanostructured materials*, 2017-2020.
3. PICT Joven 2013-1034: *Implementation of digital holography and dynamic speckle as non-destructive testing methods in the metal-mechanical industry*, 2014-2016.

As Co-director or Member of the Main Research Team

1. PICT 2022-11-00381: *Fluid dynamics in nanoporous media: capillarity, evaporation, and condensation under highly confined spatial conditions*, 2024-2027.
2. CAI+D UNL 50620190100114LI: *Films and membranes of nanostructured porous materials*, 2021-2024.
3. PIP 11220200101049CO: *Study of capillary imbibition under high spatial confinement conditions using digital holographic microscopy techniques*, 2021-2023.
4. PICT-2018-01614: *Study and characterization of hybrid perovskites for photovoltaic applications*, 2019-2022.
5. PICT-2014-1683: *Composite materials based on films and membranes of nanostructured porous materials*, 2015-2018.

As a Member of the Collaborating Group

1. CIUNSa 2924/0: *Non-destructive optical techniques applied to the 3D characterization of materials and pathogenic microorganisms*, 2024-2027.
2. PICT 2017-2397: *Digital Holographic Microscope. A tool for identification, characterization, and automatic counting of microalgae, pathogenic organisms, parasites, and particles*, 2019-2023.
3. CAI+D UNL 50120150100122LI: *Implementation and evaluation of a collaborative teaching and learning modality in a Physics course for the university's initial cycle*, 2017-2020.
4. PICT 2016-1389 and PUE 2292016010-0054: *Development and study of advanced materials and systems for solar resource exploitation*, 2017-2020.

5. Technological Linkage Project “Ing. Enrique Mosconi” (Ministry of Education and Sports): *Implementation of a solarimetric network in the province of Entre Ríos*, 2013-2015.
6. PID-UTN 25K021: *Implementation of digital holography and dynamic speckle as non-destructive testing methods*, 2013-2015.
7. PID-IN INN932: *Development and implementation of digital holography as a non-destructive testing method*, 2009-2011.

As a Fellow

1. PID-UNL 0414: *Development and study of new thin-film materials for use in photo-voltaic devices*, 2014-2017.
2. PIP 11220080101464: *Development of new technologies for the preparation of polycrystalline silicon in thin-film form suitable for solar cell production*, 2009-2013.

HUMAN RESOURCE TRAINING

Researcher Supervision

1. Supervisor of Dr. Andrea Monaldi - Assistant Researcher at CIC CONICET - 2024-present - *Topic* - Laser Optics Group, Instituto de Investigaciones en Energía No Convencional (INENCO, UNSa-CONICET), Salta.

Supervision of Ph.D. Students and Postdoctoral Researchers

1. Co-supervisor of Ph.D. Candidate Lic. Matías Sosa - 2025-present - *Mechanic and viscoelastic characterization of hydrogels by means of interferometric optical techniques* - IFIS Litoral (UNL-CONICET), Santa Fe.
2. Co-supervisor of Postdoctoral Researcher Dr. Marcelo Sallese - 24 months, 2020-2022 - *Study of capillary imbibition in nanoporous substrates using digital holographic microscopy techniques* - IFIS Litoral (UNL-CONICET), Santa Fe.
3. Co-supervisor of Ph.D. Candidate Lic. Ma. Florencia Martínez - 2019-present - *Study of capillary filling in porous materials using holographic techniques* - UNL-CONICET, Santa Fe.
4. Co-supervisor of Ph.D. in Physics Lic. Jorge Caram - 5 years, 2015-2020 - *Study of materials and hybrid devices for photovoltaic energy generation* - UNL-CONICET, Santa Fe.
5. Co-supervisor of Ph.D. in Physics Mg. Abraham Pacio Castillo - 5 years, 2013-2018 - *Thin-film solar cells from a-Si* - Benemérita Universidad Autónoma de Puebla / Instituto de Ciencias, Semiconductor Devices Research Center, Puebla, Mexico.

Supervision of Master's Students

1. Supervisor of Master's in Physics Ing. Nicolás Balducci - 5 years, 2015-2020 - *Optical characterization of thin films using holographic interferometry* - IFIS Litoral (UNL-CONICET), Santa Fe.
2. Supervisor of Master's in Physics Ing. Gabriel Gareis - 24 months, 2016-2018 - *Design and impact analysis of a grid-connected photovoltaic system: Federación case study* - National University of Rosario, Santa Fe.
3. Supervisor of Master's in Physics Ing. Juan José Stivanello - 24 months, 2016-2018 - *Study of the integration of distributed generation systems: María Grande case study* - National University of Rosario, Santa Fe.

Supervision of Research Practices of Under-Graduate Students

1. Agustín Gribaudo (Physics student, FIQ-UNL) & Teo Bragado (Industrial Engineering student, FIQ-UNL) - 2025-present - *Artificial intelligence-based weather forecasting and lightning detector development*.

2. Samuel López (Physics undergraduate student, FIQ-UNL) - 6 months, 2024 - *Weather prediction based on artificial intelligence using data from the FIQ weather station.*
3. Agustín Bogner (Physics undergraduate student, FIQ-UNL) - 6 months, 2024 - *Automatic phase aberration compensation in digital holographic microscopy.*
4. Agustina Hus (Chemical Engineering undergraduate student, FIQ-UNL) - 18 months, 2022-2023 - *Modeling and weather prediction using data from the FIQ weather station.*
5. Ma. de los Milagros Citta (Materials Engineering undergraduate student, FIQ-UNL) - 18 months, 2019-2020 - *Characterization of capillary filling dynamics in porous materials using digital holographic microscopy.*
6. Cecilia Mulone (Electronic Engineering undergraduate student, UTN Paraná) - 36 months, 2013-2016 - *Application of speckle techniques and holographic microscopy for non-destructive material testing.*
7. (Co-supervision) Matías Peralta (Materials Engineering undergraduate student, FIQ-UNL) - 6 months, 2012 - *Crystallization of thin silicon films induced by nickel.*

MANAGEMENT EXPERIENCE

1. Director of the Department of Physics (elected) at FIQ, since April 2022 (Res. CD 147/22).
2. Interim Coordinator of the Department of Physics at FIQ, from April 2021 to April 2022 (Res. CD 064/21).
3. Member of the Academic Committee for the Master's and Doctoral Programs in Physics at UNL, since April 2023 (Res. FBCB CD 321/23 and 322/23, respectively).
4. Substitute Member of the Academic Committee for the Postgraduate Program in Energy Management, FIQ-UNL, since May 2023 (Res. CD 258/23).